Thermal Stability and Fire Performance of PBS-Bagasse Sustainable Composites – Effect of DAP Treatment

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01 INTRODUCTION

The valorisation of agro-waste in bio-based polymer composites provides an environmentally friendly route to produce sustainable materials while reducing agricultural waste and the environmental impact of conventional plastics. Improved fire retardancy is key to the commercialization of agro-waste-reinforced bio-based composites. This work investigates diammonium phosphate (DAP)-treated sugarcane bagasse as a sustainable flame-retardant reinforcement for polybutylene succinate (PBS) composites.

02 OBJECTIVES

- ✓ To modify sugarcane bagasse using 10% diammonium phosphate (DAP) to improve its flame-retardant characteristics.
- ✓ To develop sustainable PBS–bagasse composites through extrusion and injection moulding
- ✓ To assess the influence of DAP treatment on the morphology, chemical characteristics, thermal stability, and fire-retardant performance of the composites

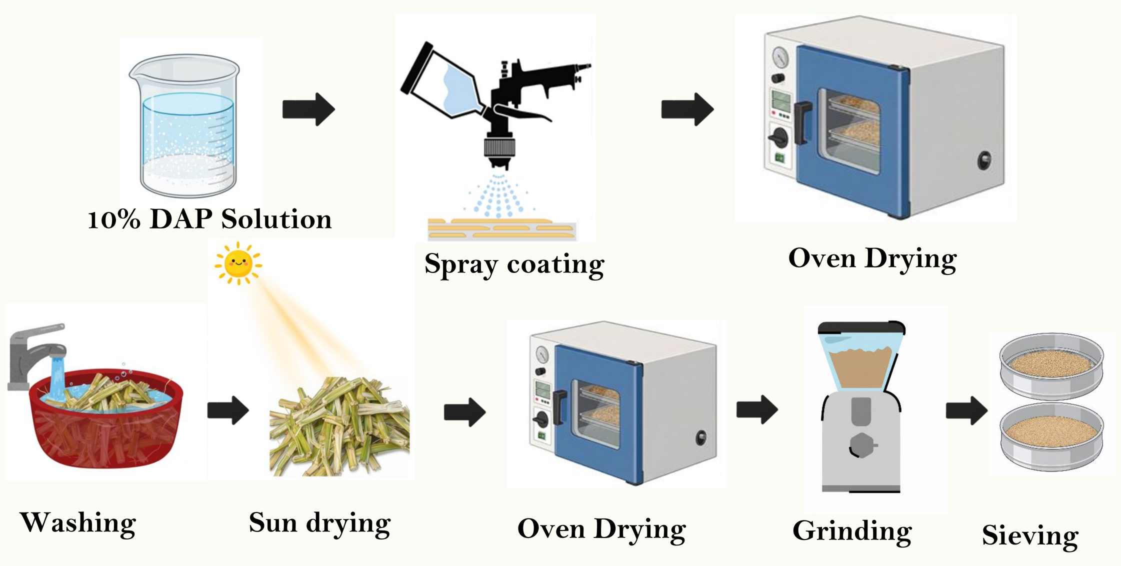
03 METHODOLOGY

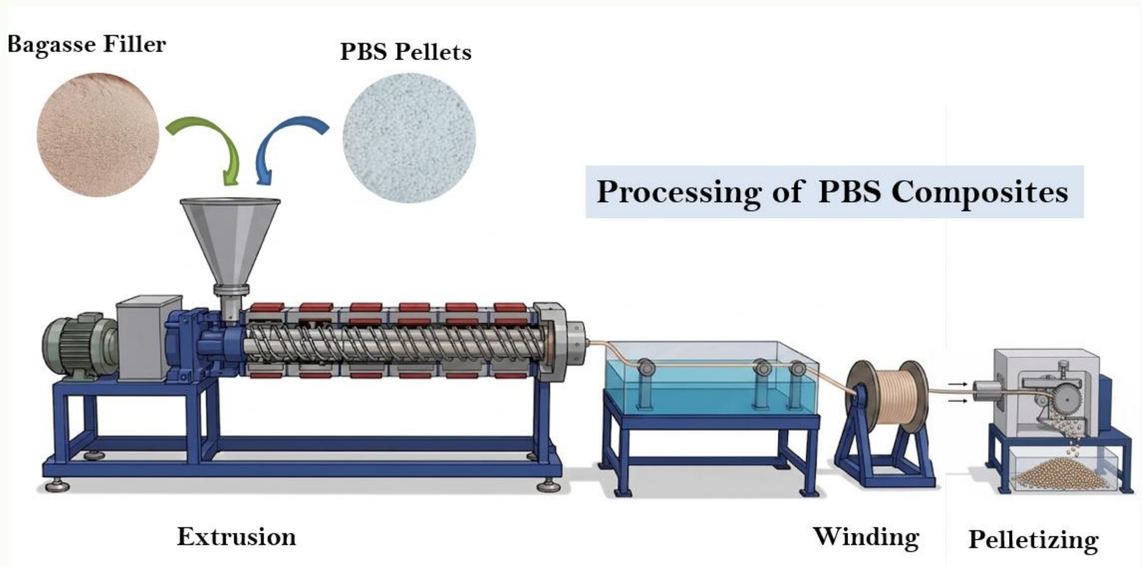
Material Preparation:

Sugarcane bagasse was washed, sun-dried, and surface-treated by spray coating with a 10 wt.% diammonium phosphate (DAP) solution. The treated bagasse was oven-dried, ground, and sieved before composite fabrication.

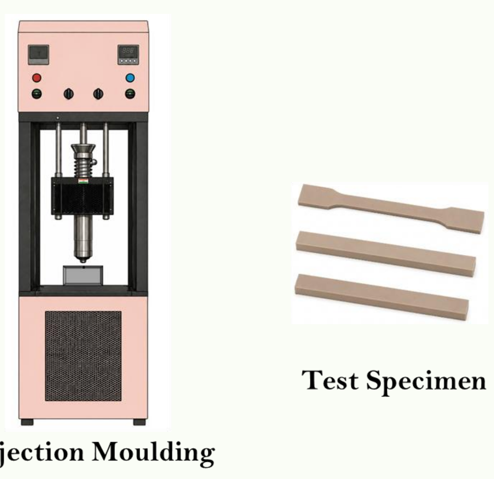
Composite Fabrication:

The treated bagasse and polybutylene succinate (PBS) pellets were melt compounded using a twin-screw extruder, pelletized, and injection moulded into standard test specimens using a semi-automatic vertical injection moulding machine.



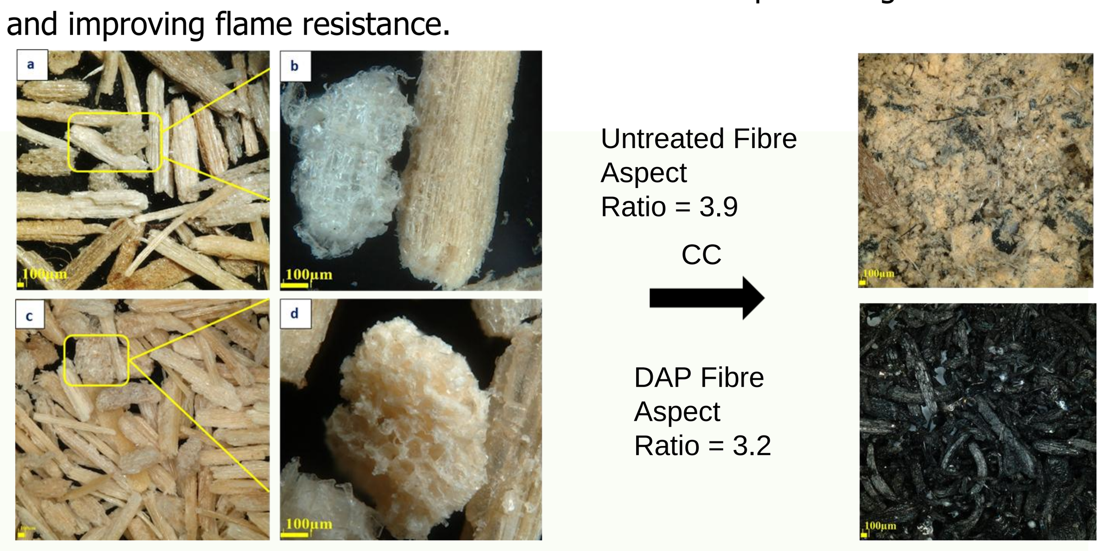


Compounding	
Number of Heating zones	8
L/D ratio	40:1
Screw Diameter (mm)	15
Die diameter (mm)	4
Screw speed (rpm)	30
PBS feeder speed	140
Bagasse feeder speed (12.5%)	9
Bagasse feeder speed (25%)	22
Winder speed (rpm)	6
Injection Molding	
Barrel Temperature	135
Mould Temperature	60

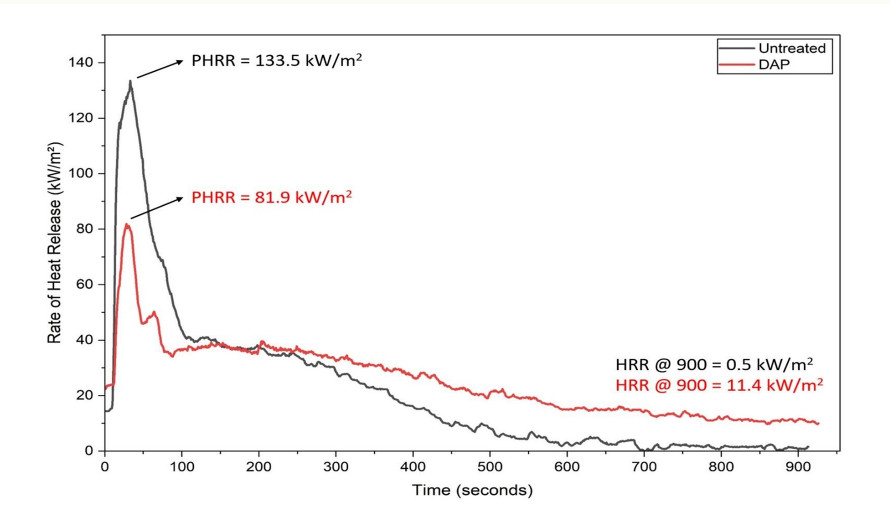


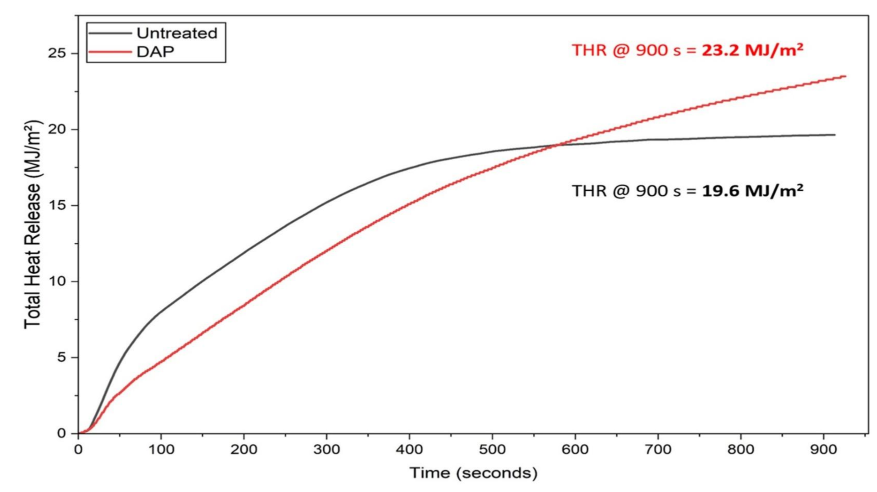
04 RESULTS

The developed PBS–bagasse composites were evaluated for their chemical, morphological, and fire-retardant performance. The results demonstrate the effectiveness of DAP treatment in promoting char formation and improving flame resistance.

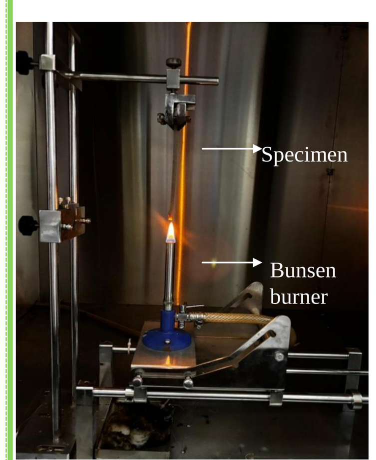


DAP treatment modified fibre surface morphology and reduced the aspect ratio through phosphate deposition and promoted the formation of a dense, continuous protective char after combustion.



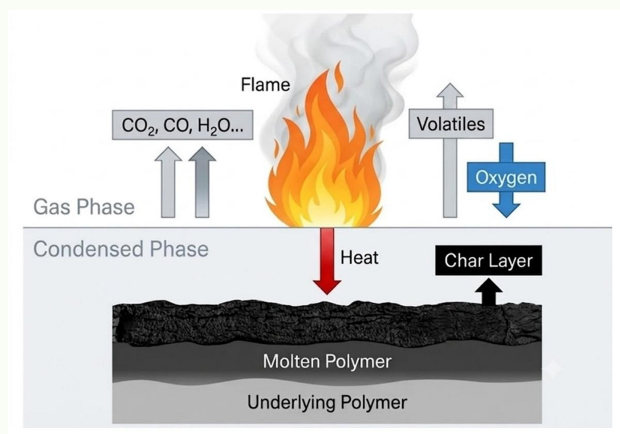


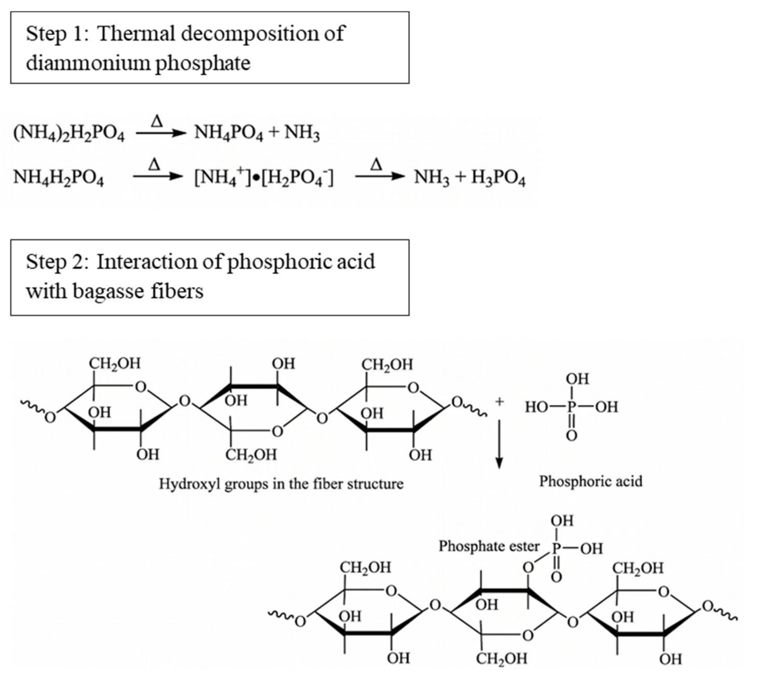
DAP treatment reduced the peak heat release rate by 38% and induced smoldering combustion, indicating improved fire resistance.



Sample	Total (s)	Dripping	Cotton Ignition	After glow	Rating
PBS	**30**	Heavy	Yes	No	**V2**
UT 12.5%	**11**	Heavy	Yes	No	**V2**
UT 25 %	**8**	Mild	Yes	No	**V2**
T 12.5 %	**11**	Mild	Yes	No	**V2**
T 25%	**2**	Mild	No	No	**V2**

Treated composites exhibited improved self-extinguishing behavior with reduced flaming drips and total flame duration

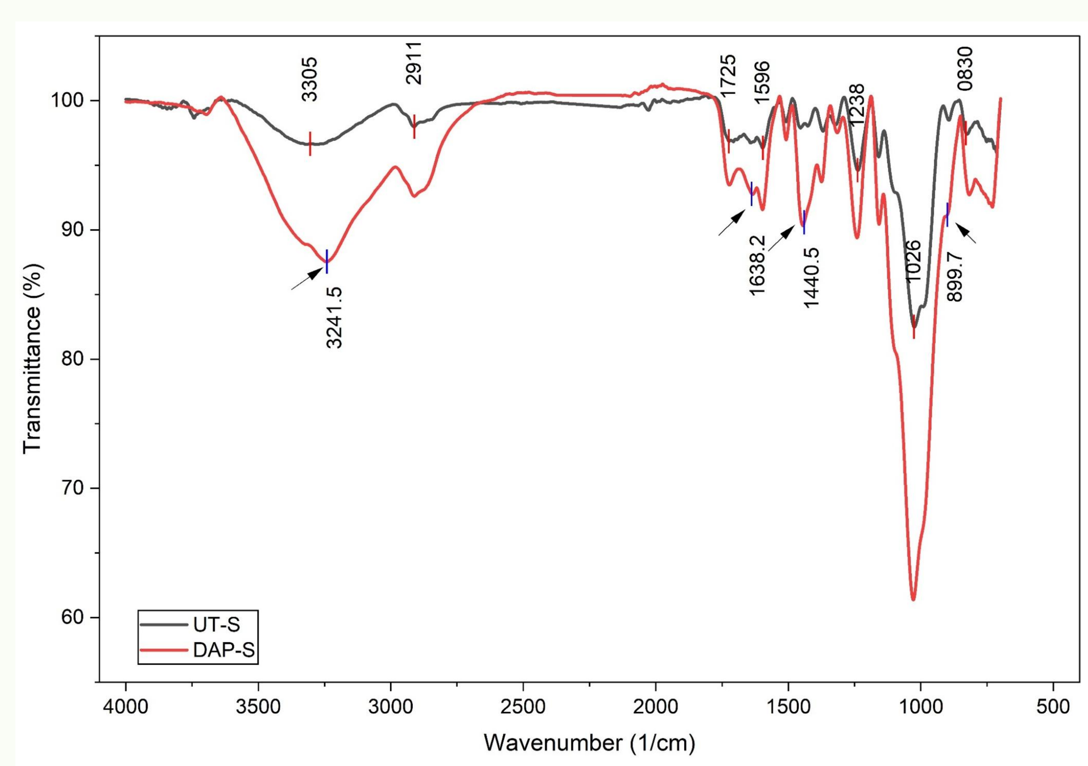




Schematic illustration of the gas-phase and condensed-phase flame-retardant mechanism of DAP-treated sugarcane bagasse/PBS composites during combustion

05 DISCUSSION

Cone calorimetry shows DAP treatment lowers the peak heat release rate by roughly 35-40% relative to the untreated composite and slows the rate of heat release, consistent with early formation of a protective char layer. The mechanism is supported by the FTIR spectrum of the DAP treated Bagasse.



Wavenumber (1/cm) 10% DAP Treated	Functional group	Description
3241	N–H Stretching	From ammonia (NH_3)
1638	N–H bending	From ammonia (NH_3)
1440	P=O Stretching	Phosphorus species
899	P=O Stretching	Phosphorus ester

- ❖ FTIR analysis confirmed the incorporation of phosphate species on the bagasse surface by the appearance of characteristic P=O and NH vibrations, indicating successful DAP deposition.
- ❖ During combustion, DAP decomposes to phosphoric acid and polyphosphoric acid, which catalyze cellulose dehydration and promote the formation of a stable carbonaceous char layer.
- ❖ Consequently, the treated composites exhibited approximately 38% lower peak heat release rate together with improved self-extinguishing behavior and reduced flaming drips during UL-94 testing.
- ❖ The protective char layer acts as a thermal and mass-transfer barrier, limiting heat feedback and volatile release, thereby slowing combustion and enhancing fire performance without compromising the sustainability of the bio-composite.

06 CONCLUSIONS

- The DAP treatment reduced the PHRR and improved self-extinguishing properties in the composites.
- The FTIR revealed the phosphorous species in the treated Bagasse.
- The developed composite is safe, halogen free and fits in the circular economy

The chemical interaction between DAP and Bagasse drives early protective char formation, reducing the peak heat release rate by 38%. Upgrading the highly sustainable agricultural waste into a functional fire-retardant filler, which is incorporated into a Bioplastic presents a highly sustainable alternative to traditional plastics supporting the circular plastic economy.

07 FUTURE WORK

To optimize DAP loading concentration to balance mechanical and flame-retardant properties. Then to further Investigate long-term weathering and biodegradation behavior of the Bio-composites and explore it's application.

08 REFERENCES

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